

Exercise Sheet 2 Solutions

Exercise 2.1 Induction.

- (a) Prove via mathematical induction that for all integers $n \geq 5$,

$$2^n > n^2.$$

In your solution, you should address the base case, the induction hypothesis and the induction step.

- (b) Let x be any real number. Prove via mathematical induction that for every positive integer n , we have

$$(1+x)^n = \sum_{i=0}^n \binom{n}{i} x^i,$$

where

$$\binom{n}{i} = \frac{n!}{i!(n-i)!}.$$

Recall that the factorial of a positive integer n is defined as $n! = 1 \cdot 2 \cdot \dots \cdot (n-1) \cdot n$. We use a standard convention $0! = 1$, so $\binom{n}{0} = \binom{n}{n} = 1$ for every positive integer n .

In your solution, you should address the base case, the induction hypothesis and the induction step.

Hint: You can use the following fact without proof: for every $1 \leq i \leq n$,

$$\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}.$$

Solution:

Base case: $n=5$

$$2^5 = 32 > 25 = 5^2$$

$\Rightarrow$ the statement holds for $n=5$.

Induction Hypothesis: We assume the statement holds for $n=k$,

$$k \in \mathbb{N}, k \geq 5$$

$$\Rightarrow 2^k > k^2$$

Induction Step: We will show the statement holds for $n=k+1$

We will show $2^{k+1} > (k+1)^2$

$$2^{k+1} = 2 \cdot 2^k$$

$$> 2 \cdot k^2 \text{ (from IH we know } 2^k > k^2 \text{)}$$

$$= k^2 + k^2$$

$$\geq k^2 + 5k \text{ (we know } k \geq 5 \text{)}$$

$$= k^2 + 2k + 3k$$

$$\geq k^2 + 2k + 15 \text{ (we know } k \geq 5 \text{)}$$

$$> k^2 + 2k + 1$$

$$= (k+1)^2$$

$\Rightarrow$ We have shown $2^{k+1} > (k+1)^2$

$\Rightarrow$ The statement holds for $n = k+1$

$\Rightarrow$ By the principle of mathematical induction, the statement holds for $\forall n \in \mathbb{N}$, $n \geq 5$

B) Base case: $n = 1$

$$(1+x)^1 = 1+x = 1 \cdot x^0 + 1 \cdot x^1 = \binom{1}{0} x^0 + \binom{1}{1} x^1 = \sum_{i=0}^1 \binom{1}{i} x^i$$

$\Rightarrow$ the statement holds for $n = 1$.

Induction hypothesis: We assume the statement holds for $n = k$, $k \in \mathbb{N}^+$

$$\Rightarrow (1+x)^k = \sum_{i=0}^k \binom{k}{i} x^i$$

Induction step: We will show the statement holds for $n = k+1$

We must show: $(1+x)^{k+1} = \sum_{i=0}^{k+1} \binom{k+1}{i} x^i$

$$(1+x)^{k+1} = (1+x)^k (1+x)$$

$$\stackrel{IH}{=} \sum_{i=0}^k \binom{k}{i} x^i \cdot (1+x)$$

$$= \sum_{i=0}^k \binom{k}{i} x^i + \sum_{i=0}^k \binom{k}{i} x^{i+1}$$

$$= \sum_{i=0}^k \binom{k}{i} x^i + \sum_{i=1}^{k+1} \binom{k}{i-1} x^i$$

$$= \binom{k}{0} \cdot x^0 + \sum_{i=1}^k \binom{k}{i} x^i + \sum_{i=1}^k \binom{k}{i-1} x^i + \binom{k}{k} x^{k+1}$$

$$= 1 \cdot 1 + \sum_{i=1}^k \binom{k}{i} x^i + \sum_{i=1}^k \binom{k}{i-1} x^i + 1 \cdot x^{k+1}$$

$$= 1 + \sum_{i=1}^k \left(\binom{k}{i} + \binom{k}{i-1} \right) x^i + 1 \cdot x^{k+1}$$

$$= 1 + \sum_{i=1}^k \binom{k+1}{i} x^i + 1 \cdot x^{k+1}$$

$$= 1 + \sum_{i=1}^k \binom{k+1}{i} x^i + 1 \cdot x^{k+1}$$

$$= \binom{k+1}{0} x^0 + \sum_{i=1}^k \binom{k+1}{i} x^i + \binom{k+1}{k+1} x^{k+1}$$

$$= \sum_{i=0}^{k+1} \binom{k+1}{i} x^i$$

$\Rightarrow$ We have shown that:

$$(1+x)^{k+1} = \sum_{i=0}^{k+1} \binom{k+1}{i} x^i$$

$\Rightarrow$ We have shown that the statement holds for $n = k+1$

$\Rightarrow$ By the principle of mathematical induction, the statement holds for $\forall n \in \mathbb{N}^+$

Comment: The solution for part a) is the same as the

master solution, the solution for part 8) is very similar, in my solution the long equality is written in great detail in order to show where each summand comes from.

This solution has the appropriate level of detail and formalism for the exam.

Exercise 2.2 Fibonacci numbers (1 point).

There are a lot of neat properties of the Fibonacci numbers that can be proved by induction. Recall that the Fibonacci numbers are defined by $f_0 = 0$, $f_1 = 1$ and the recursion relation $f_{n+1} = f_n + f_{n-1}$ for all $n \geq 1$. For example, $f_2 = 1$, $f_5 = 5$, $f_{10} = 55$, $f_{15} = 610$.

Prove that $f_n \geq \frac{1}{3} \cdot 1.5^n$ for $n \geq 1$.

In your solution, you should address the base case, the induction hypothesis and the induction step.

Solution:

Base case: $n = 1$

9. 0,25

$$f_1 = 1 \geq 0,5 = \frac{1}{3} \cdot 1,5^1 = \frac{1}{3} \cdot 1,5^n$$

$\Rightarrow$ the statement holds for $n = 1$

$n = 2$

$$f_2 = 1 \geq 0,75 = \frac{1}{3} \cdot 1,5^2 = \frac{1}{3} \cdot 1,5^n$$

$\Rightarrow$ the statement holds for $n = 2$

Hypothesis: We assume the statement holds for all $n \leq k$, $k \geq 2$ (*)
 $\Rightarrow f_n \geq \frac{1}{3} \cdot 1,5^n \quad \forall n \leq k$

Induction step: We will show the statement holds for $n = k+1$

We must show that $f_{k+1} \geq \frac{1}{3} \cdot 1,5^{k+1}$

$$\begin{aligned} f_{k+1} &= f_{k-1} + f_k \quad (*) \\ &\stackrel{IH}{\geq} \frac{1}{3} \cdot 1,5^{k-1} + \frac{1}{3} \cdot 1,5^k \\ &= \frac{1}{3} \cdot 1,5^{k-1} (1 + 1,5) \\ &= \frac{1}{3} \cdot 1,5^{k-1} \cdot 2,5 \\ &> \frac{1}{3} \cdot 1,5^{k-1} \cdot 2,25 \\ &= \frac{1}{3} \cdot 1,5^{k-1} \cdot 1,5^2 \\ &= \frac{1}{3} \cdot 1,5^{k+1} \end{aligned}$$

$\Rightarrow$ We have shown $f_{k+1} > \frac{1}{3} \cdot 1,5^{k+1}$

$\Rightarrow$ The statement holds for $n = k+1$

$\Rightarrow$ By the principle of mathematical induction, the statement holds for $\forall n \in \mathbb{N}^+$

(*) Be careful to specify $k \geq 2$ in the induction hypothesis.

We can do that because in the base case we have seen that the statement holds for $n=1$ and $n=2$, so by making $k \geq 2$, we still will have shown the statement holds for $n=1$.

Why $k \geq 2$? See (*) in the induction step. If k was ≤ 1 , then $p_{k-1} = p_0$ and for $p_0 = 0$ it does not hold $p_0 \geq \frac{1}{3} \cdot 1,5^0$
 $\Rightarrow$ we cannot use it to write inequality 1.

If in your solution is the same as mine and you say $k \geq 1$ in the hypothesis, you would need to explain the case where $k=1$ separately in the induction step.

Comment: My solution uses strong induction. The master solution is a little different, when it comes to the induction hypothesis. This solution demonstrates an appropriate level of detail and formalism for the exam.

Exercise 2.3 *O*-notation quiz (1 point).

(a) For all the following functions the variable n ranges over $\mathbb{N}$. Prove or disprove the following statements. Justify your answer using Theorems 1 and/or 2.

(1) $2n^5 + 10n^2 \leq O(\frac{1}{100}n^6)$

(2) $n^{10} + 2n^2 + 7 \leq O(100n^9)$

(3) $e^{1.2n} \leq O(e^n)$

(4)* $n^{\frac{2n+3}{n+1}} \leq O(n^2)$

(b) This question does not count toward the bonus point. Find f and g as in Theorem 1 such that $f \leq O(g)$, but the limit $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$ does not exist. This proves that the first point of Theorem 1 provides a sufficient, but not a necessary condition for $f \leq O(g)$.

Solution:

a) (1) We will prove the statement using Theorem 1.

$$\lim_{n \rightarrow \infty} \frac{2n^5 + 10n^2}{\frac{1}{100}n^6} = \lim_{n \rightarrow \infty} \frac{200n^5}{n^6} + \frac{1000n^2}{n^6} = \lim_{n \rightarrow \infty} \frac{200}{n} + \frac{1000}{n^4} = 0$$

$\Rightarrow 2n^5 + 10n^2 \leq O(\frac{1}{100}n^6)$ by Th. 1 and the statement is true

b) We will disprove this statement using Theorem 1

$$\lim_{n \rightarrow \infty} \frac{n^{10} + 2n^2 + 7}{100n^9} = \lim_{n \rightarrow \infty} \frac{n^{10}}{100n^9} + \frac{2n^2}{100n^9} + \frac{7}{100n^9} =$$

$$\lim_{n \rightarrow \infty} \frac{n}{100} + \frac{1}{50n^7} + \frac{7}{100n^9} = \lim_{n \rightarrow \infty} \frac{n}{100} = \infty$$

$\Rightarrow n^{10} + 2n^2 + 7 \neq O(100n^9)$ by Th.1 and the statement is false

(3) We will disprove the the statement using Theorem 1.

$$\lim_{n \rightarrow \infty} \frac{e^{1.2n}}{e^n} = \lim_{n \rightarrow \infty} e^{1.2n - n} = \lim_{n \rightarrow \infty} e^{0.2n} = \infty$$

$\Rightarrow e^{1.2n} \neq O(e^n)$ by Th.1 and the statement is false

(4) We will prove the statement using Theorem 1.

$$\lim_{n \rightarrow \infty} \frac{n^{\frac{2n+3}{n+1}}}{n^2} = \lim_{n \rightarrow \infty} \frac{e^{\ln n^{\frac{2n+3}{n+1}}}}{e^{\ln n^2}} = \lim_{n \rightarrow \infty} \frac{e^{\frac{2n+3}{n+1} \cdot \ln n}}{e^{2 \ln n}} =$$

$$\lim_{n \rightarrow \infty} e^{\frac{2n+3}{n+1} \cdot \ln n - 2 \ln n} = \lim_{n \rightarrow \infty} e^{\ln n \left(\frac{2n+3}{n+1} - 2 \right)} =$$

$$e^{\lim_{n \rightarrow \infty} \ln n \left(\frac{2n+3}{n+1} - 2 \right)}$$

$$\lim_{n \rightarrow \infty} \ln n \left(\frac{2n+3}{n+1} - 2 \right) = \lim_{n \rightarrow \infty} \ln n \left(\frac{2(n+1)+1}{n+1} - 2 \right) =$$

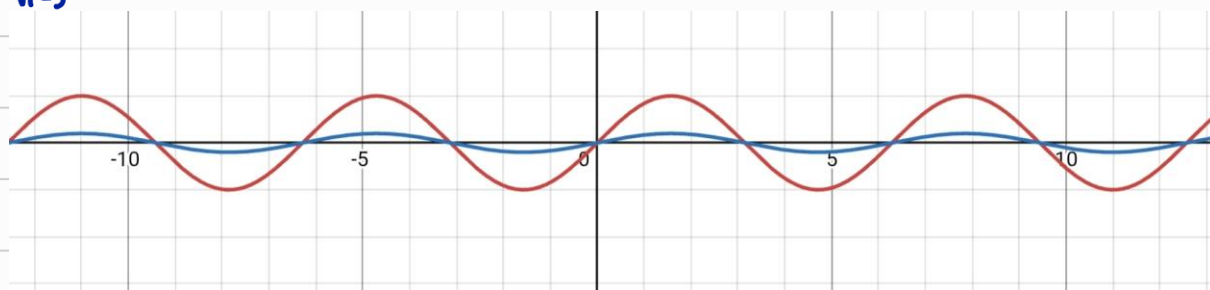
$$\lim_{n \rightarrow \infty} \ln n \left(2 + \frac{1}{n+1} - 2 \right) = \lim_{n \rightarrow \infty} \ln n \cdot \frac{1}{n+1} = \lim_{n \rightarrow \infty} \frac{\ln n}{n+1} = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^{\frac{2n+3}{n+1}}}{n^2} = e^{\lim_{n \rightarrow \infty} \ln n \left(\frac{2n+3}{n+1} - 2 \right)} = e^0 = 1$$

$\Rightarrow n^{\frac{2n+3}{n+1}} \leq O(n^2)$ by Th 1 and the statement is true

(b) We define $f(x) := \sin(x)$, $g(x) = 5$.

$\lim_{n \rightarrow \infty} \frac{\sin(x)}{5}$ does not exist



$$f(x) = \sin(x)$$

$$g(x) = \frac{\sin(x)}{5}$$

Comment: My example for part b) is different from the master solution. My solution for part a) (4) is different than the master solution and is written in greater detail than what is expected of you. Part a) 1, 2, 3 demonstrate the appropriate level of detail and formalism for the exam.

Exercise 2.4 Asymptotic growth of $\sum_{i=1}^n \frac{1}{i}$ (1 point).

The goal of this exercise is to show that the sum $\sum_{i=1}^n \frac{1}{i}$ behaves, up to constant factors, as $\log(n)$ when n is large. Formally, we will show $\sum_{i=1}^n \frac{1}{i} \leq O(\log n)$ and $\log n \leq O(\sum_{i=1}^n \frac{1}{i})$ as functions from $\mathbb{N}_{\geq 2}$ to $\mathbb{R}^+$.

For parts (a) to (c) we assume that $n = 2^k$ is a power of 2 for $k \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}$. We will generalise the result to arbitrary $n \in \mathbb{N}$ in part (d). For $j \in \mathbb{N}$, define

$$S_j = \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{i}.$$

(a) For any $j \in \mathbb{N}$, prove that $S_j \leq 1$.

Hint: Find a common upper bound for all terms in the sum and count the number of terms.

Solution:

$$\begin{aligned} \text{a) } S_j &= \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{i}, \text{ } j \text{ is arbitrary} \\ &\Rightarrow i \geq 2^{j-1}+1 \\ &\Rightarrow \frac{1}{i} \leq \frac{1}{2^{j-1}+1} \end{aligned}$$

Additionally, we know that i iterates over all integers between $2^{j-1}+1$ and 2^j , inclusive

$\Rightarrow$ the sum $\sum_{i=2^{j-1}+1}^{2^j} \frac{1}{i}$ has:

$$2^j - (2^{j-1}+1) + 1 = 2^j - 2^{j-1} = 2 \cdot 2^{j-1} - 2^{j-1} = 2^{j-1} \text{ terms}$$

$\Rightarrow$ We know that the sum has 2^{j-1} terms and for each term $\frac{1}{i}$ holds $\frac{1}{i} \leq \frac{1}{2^{j-1}+1}$

$$\Rightarrow S_j = \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{i} \leq \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{2^{j-1}+1} = 2^{j-1} \cdot \frac{1}{2^{j-1}+1} = \frac{2^{j-1}}{2^{j-1}+1} < 1$$

$\Rightarrow S_j \leq 1$ and since j is arbitrary,
 $S_j \leq 1 \quad \forall j \in \mathbb{N}$

(b) For any $j \in \mathbb{N}$, prove that $S_j \geq \frac{1}{2}$.

$$\text{b) } S_j = \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{i}, \text{ } j \text{ is arbitrary}$$

$$\Rightarrow i \leq 2^j$$

$$\Rightarrow \frac{1}{i} \geq \frac{1}{2^j}$$

$$\Rightarrow S_j = \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{i} \geq \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{2^j}$$

The sum $\sum_{i=2^{j-1}+1}^{2^j}$ has $2^j - (2^{j-1}+1) + 1 = 2 \cdot 2^{j-1} - 2^{j-1} = 2^{j-1}$ terms

$$\Rightarrow S_j = \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{i} \geq \sum_{i=2^{j-1}+1}^{2^j} \frac{1}{2^j} = 2^{j-1} \cdot \frac{1}{2^j} = \frac{1}{2}$$

$$\Rightarrow S_j \geq \frac{1}{2} \text{ and since } j \text{ is arbitrary } S_j \geq \frac{1}{2} \forall j \in \mathbb{N}$$

c) Let $k \in \mathbb{N}_0$ be arbitrary

$$\sum_{i=1}^{2^k} \frac{1}{i} = 1 + \sum_{i=2^0+1}^{2^1} \frac{1}{i} + \sum_{i=2^1+1}^{2^2} \frac{1}{i} + \dots + \sum_{i=2^{k-1}+1}^{2^k} \frac{1}{i}$$

$$= 1 + \sum_{j=1}^k S_j$$

From part a) we know that $S_j \leq 1 \forall j \in \mathbb{N}$

$$\Rightarrow \sum_{i=1}^{2^k} \frac{1}{i} = 1 + \sum_{j=1}^k S_j \leq 1 + k \cdot 1 = 1 + k$$

$$\Rightarrow \sum_{i=1}^{2^k} \frac{1}{i} \leq k+1 \text{ and since } k \text{ is arbitrary } \sum_{i=1}^{2^k} \frac{1}{i} \leq k+1 \forall k \in \mathbb{N}_0$$

From part b) we know that $S_j \geq \frac{1}{2} \forall j \in \mathbb{N}$

$$\Rightarrow \sum_{i=1}^{2^k} \frac{1}{i} = 1 + \sum_{j=1}^k S_j \geq 1 + k \cdot \frac{1}{2} > \frac{k+1}{2}$$

$$\Rightarrow \sum_{i=1}^{2^k} \frac{1}{i} \geq \frac{k+1}{2} \text{ and since } k \text{ is arbitrary } \sum_{i=1}^{2^k} \frac{1}{i} \geq \frac{k+1}{2} \forall k \in \mathbb{N}_0$$

d) Let n be arbitrary and $k_1 = \lceil \log_2 n \rceil$, $k_2 = \lfloor \log_2 n \rfloor$

$$\Rightarrow \sum_{i=1}^n \frac{1}{i} \leq \sum_{i=1}^{2^{k_1}} \frac{1}{i} \text{ and } \sum_{i=1}^n \frac{1}{i} \geq \sum_{i=1}^{2^{k_2}} \frac{1}{i}$$

From part c) we know that $\sum_{i=1}^{2^k} \frac{1}{i} \leq k+1 \forall k \in \mathbb{N}_0$, including k_1

$$\Rightarrow \sum_{i=1}^{2^{k_1}} \frac{1}{i} \leq k_1 + 1$$

$$\text{We know } k_1 = \lceil \log_2 n \rceil \Rightarrow k_1 \leq \log_2 n + 1$$

$$\Rightarrow \sum_{i=1}^n \frac{1}{i} \leq \sum_{i=1}^{2^{k_1}} \frac{1}{i} \leq k_1 + 1 \leq \log_2 n + 1 + 1$$

$$\Rightarrow \sum_{i=1}^n \frac{1}{i} \leq \log_2 n + 2 \text{ and since } n \text{ is arbitrary,}$$

$$\sum_{i=1}^n \frac{1}{i} \leq \log_2 n + 2 \quad \forall n \in \mathbb{N}$$

From part c) we know that $\sum_{i=1}^{2^k} \frac{1}{i} \geq \frac{k+1}{2} \quad \forall k \in \mathbb{N}_0$, including k_2

$$\Rightarrow \sum_{i=1}^{2^{k_2}} \frac{1}{i} \geq \frac{k_2 + 1}{2}$$

We know $k_2 = \lfloor \log_2 n \rfloor \Rightarrow k_2 \geq \log_2 n - 1$

$$\Rightarrow \sum_{i=1}^n \frac{1}{i} \geq \sum_{i=1}^{2^{k_2}} \frac{1}{i} \geq \frac{k_2 + 1}{2} \geq \frac{\log_2 n - 1 + 1}{2} = \frac{\log_2 n}{2}$$

$$\Rightarrow \sum_{i=1}^n \frac{1}{i} \geq \frac{\log_2 n}{2} \text{ and since } n \text{ is arbitrary}$$

$$\sum_{i=1}^n \frac{1}{i} \geq \frac{\log_2 n}{2} \quad \forall n \in \mathbb{N}$$

Comment: My solution is very similar to the master solution. In part c) I have included explanation of why $\sum_{i=1}^{2^k} \frac{1}{i} = 1 + \sum_{j=1}^k S_j$. You do not need to include that in your solutions. This solution demonstrates an appropriate level of detail and formalism.

Exercise 2.5 Asymptotic growth of $\ln(n!)$.

Recall that the factorial of a positive integer n is defined as $n! = 1 \cdot 2 \cdot \dots \cdot (n-1) \cdot n$. For the following functions n ranges over $\mathbb{N}_{\geq 2}$.

(a) Show that $\ln(n!) \leq O(n \ln n)$.

Hint: You can use the fact that $n! \leq n^n$ for $n \in \mathbb{N}_{\geq 2}$ without proof.

(b) Show that $n \ln n \leq O(\ln(n!))$.

Hint: You can use the fact that $\left(\frac{n}{2}\right)^{\frac{n}{2}} \leq n!$ for $n \in \mathbb{N}_{\geq 2}$ without proof.

Solution:

a) We know that $\ln n$ is monotonically increasing for $n \in \mathbb{N}_{\geq 2}$
 $\Rightarrow$ since $n! \leq n^n \quad \forall n \in \mathbb{N}_{\geq 2}$, then

$$\ln(n!) \leq \ln(n^n) \quad \forall n \in \mathbb{N}_{\geq 2}$$

$$\Rightarrow \ln(n!) \leq n \ln n \quad \forall n \in \mathbb{N}_{\geq 2}$$

$\Rightarrow \exists C > 0$ (namely $C=1$) $\forall n \in \mathbb{N}_{\geq 2}$, such that $\ln(n!) \leq C \cdot n \ln n$

$\Rightarrow \ln(n!) \leq O(n \ln n)$ By Definition 1.

b) We know that $(\frac{n}{2})^{\frac{n}{2}} \leq n! \quad \forall n \in \mathbb{N}_{\geq 2}$ and $\ln n$ is monotonically increasing for $n \in \mathbb{N}_{\geq 2}$

$$\Rightarrow \ln\left(\left(\frac{n}{2}\right)^{\frac{n}{2}}\right) \leq \ln(n!)$$

$$\Rightarrow \frac{n}{2} \ln\left(\frac{n}{2}\right) \leq \ln(n!)$$

$$\Rightarrow n \ln\left(\frac{n}{2}\right) \leq 2 \ln(n!)$$

$$\Rightarrow n(\ln n - \ln 2) \leq 2 \ln(n!)$$

$$\Rightarrow n \ln n - n \ln 2 \leq 2 \ln(n!)$$

$$\Rightarrow n \ln n \leq 2 \ln(n!) + n \ln 2$$

$$n \ln 2 = \sum_{i=1}^n \ln 2 = \ln 2 + \sum_{i=2}^n \ln 2 \leq \ln 2 + \sum_{i=2}^n \ln n = \ln 2 + \ln(n!)$$

$$\Rightarrow n \ln 2 \leq \ln 2 + \ln(n!)$$

$$\text{and } \ln 2 \leq \ln(n!) \quad (n \in \mathbb{N}_{\geq 2})$$

$$\Rightarrow n \ln 2 \leq 2 \ln(n!)$$

$$\Rightarrow n \ln n \leq 2 \ln(n!) + n \ln 2 \leq 2 \ln(n!) + 2 \ln(n!) = 4 \ln(n!)$$

$$\Rightarrow n \ln n \leq 4 \ln(n!)$$

$$\Rightarrow \exists C > 0 \text{ (namely } C=4) \forall n \in \mathbb{N}_{\geq 2}, \text{ such that } n \ln n \leq C \cdot \ln(n!)$$

$$\Rightarrow n \ln n \leq O(\ln(n!))$$

Comment: This solution is the same as the master solution.

The conclusions in part a) and part b) as well as the calculations on part b) are a little more detailed in this solution to help with understanding.

Moodle Quiz

Frage 1

Richtig

Erreichte Punkte
0,20 von 0,20

Frage
markieren

$$n^8 - 800n^7 - 10000n^6 + 200n^2 \leq O(0,25n^7)$$

☐ Wahr

☒ Falsch

Solution: We will use Theorem 1 to disprove the statement

$$\lim_{n \rightarrow \infty} \frac{n^8 - 800n^7 - 10000n^6 + 200n^2}{0,25n^7} = \lim_{n \rightarrow \infty} \frac{n^8}{0,25n^7} - \frac{800n^7}{0,25n^7} - \frac{10000n^6}{0,25n^7} + \frac{200n^2}{0,25n^7} =$$

$$\lim_{n \rightarrow \infty} \frac{n}{0,25} - 800 - \frac{10000}{0,25n} + \frac{200}{0,25n^5} = \lim_{n \rightarrow \infty} \frac{n}{0,25} = \infty$$

$$\Rightarrow n^8 - 800n^7 - 10000n^6 + 200n^2 \notin O(0,25n^7)$$

Let $f(n) = 4e^n$, for which of the following definitions of $g(n)$ is $f(n) \leq O(g(n))$?

True	False			
☐	☒	$g(n) = n^4 + n^2$	☒	$g(n) = n^4 + n^2$ is polynomial, so it grows slower than any exponential, in particular slower than $f(n)$, so $f(n) \not\leq O(g(n))$.
☐	☒	$g(n) = n^{100} + e^{100 \ln(n)}$	☒	$e^{100 \ln(n)} = n^{100}$ so $g(n) = 2n^{100}$ and this grows slower than any exponential, in particular slower than $f(n)$ so $f(n) \not\leq O(g(n))$.
☒	☐	$g(n) = 2^{3n}$	☒	$2^{3n} = 8^n = e^{\ln(8)n}$, so we can see the limit of the ratio $\frac{f(n)}{g(n)} = \frac{4e^n}{e^{\ln(8)n}} = \frac{4}{e^{(\ln(8)-1)n}}$ goes to zero as $\ln(8) > \ln(e) = 1$, meaning $f(n) \leq O(g(n))$

Bewertung: Teilpunkte 2

Solution: We will use theorem 1.

$$1) \lim_{n \rightarrow \infty} \frac{4e^n}{n^4 + n^2} = \lim_{n \rightarrow \infty} \frac{e^{\ln 4} \cdot e^n}{e^{\ln n^4 + n^2}} = \lim_{n \rightarrow \infty} \frac{e^{\ln 4 + n}}{e^{\ln n^4 + n^2}} = \lim_{n \rightarrow \infty} e^{\ln 4 + n - \ln n^4 - n^2} = e^{\lim_{n \rightarrow \infty} \ln 4 + n - 6 \ln n} = e^\infty = \infty \Rightarrow \text{the statement is false}$$

! Polynomial functions always grow asymptotically slower than exponential functions.

Polynomial function example:

$p(x) = 10x^5 + 8x^4 + 3x^2 + 5 \rightarrow$ the argument is the base

Exponential function example:

$q(x) = 21^x \rightarrow$ the argument is the power

2) $g(n) = n^{100} + e^{100 \ln(n)} = n^{100} + n^{100} = 2 \cdot n^{100} \Rightarrow g(n)$ is a polynomial function, while $f(n)$ is an exponential function

$\Rightarrow f(n) \not\leq O(g(n))$

$$3) \lim_{n \rightarrow \infty} \frac{4e^n}{2^{3n}} = \lim_{n \rightarrow \infty} \frac{e^{\ln 4} \cdot e^n}{e^{3n \ln 2}} = \lim_{n \rightarrow \infty} e^{n + \ln 4 - 3n \ln 2} = e^{\lim_{n \rightarrow \infty} n + \ln 4 - 3n \ln 2} = e^{\lim_{n \rightarrow \infty} n(1 - 3 \ln 2) + \ln 4} = e^{-\infty} = 0$$

$\Rightarrow f(n) \leq O(g(n))$

You want to show using induction that a statement $A(n)$ holds for all n such that $n \geq 2$.

Which of the following combinations of *base case* and *induction step* would form a valid proof?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Base case: $A(2)$ holds. ☒ This is just standard induction.
Induction step: $A(k) \Rightarrow A(k+1)$ for all integers $k \geq 2$
- ☒ b. Base case: $A(2), A(3)$ holds. ☒ From base case $n = 2$ we prove it for all positive even numbers and from base case $n = 3$ we prove it for all numbers at least 3, so this covers all the numbers.
Induction step: $A(k) \Rightarrow A(k+2)$ for all integers $k \geq 2$.
- ☐ c. Base case: $A(2)$ holds.
Induction step: $A(k) \Rightarrow A(k+2)$ for all integers $k \geq 2$.

Solution: a) This is normal induction

b) We have shown $A(2)$ and $A(3)$ hold and the induction step shows that if $A(k)$ holds then $A(k+2)$ holds

$\Rightarrow$ from A(2), we can show that A(2r) holds $\forall r \geq 1$ and
 from A(3), we can show that A(2r+1) holds $\forall r \geq 1$
 $\Rightarrow$ A(n) holds $\forall n \geq 2$

c) Because the base case is missing A(3) we can only show that
 A(n) holds for the even numbers $n \in \mathbb{N} \geq 2$

Frage 4
 Richtig

Erreichte Punkte
 0.20 von 0.20

Frage
 markieren

$\sum_{i=1}^n i^2 \leq O(n^2 \log(n))$?

- ☐ Wahr
☒ Falsch

Solution: $\sum_{i=1}^n i^2 = \sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} i^2 + \sum_{i=\lfloor \frac{n}{2} \rfloor + 1}^n i^2 \geq \sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} 1^2 + \sum_{i=\lfloor \frac{n}{2} \rfloor + 1}^n \left(\frac{n}{2}\right)^2 =$
 $\lfloor \frac{n}{2} \rfloor \cdot 1^2 + (n - \lfloor \frac{n}{2} \rfloor + 1) \left(\frac{n}{2}\right)^2 \geq \left(\frac{n}{2} - 1\right) \cdot 1^2 + \frac{n}{2} \cdot \left(\frac{n}{2}\right)^2 = \frac{n}{2} - 1 + \frac{n^3}{2} \geq \frac{n^3}{2}$
 $\Rightarrow \sum_{i=1}^n i^2 \geq \frac{n^3}{2}$
 $\Rightarrow \sum_{i=1}^n i^2 \neq O(n^2 \log(n))$
 (for $n \geq 2$)

Frage 5

Richtig

Erreichte Punkte
 0.20 von 0.20

Frage
 markieren

Frage
 bearbeiten

v3 (neueste)

Consider the following recursive formula:

$$T(1) = 0$$

$$T(n) = 2T(n/2) + 5n$$

Assume n is of the form $n = 2^k$ for $k \in \{0, 1, 2, 3, \dots\}$.

Is $T(n) \leq O(n)$?

- ☐ Wahr
☒ Falsch

Solution:

$$T(1) = 0$$

$$\begin{aligned} T(n) &= 2T(n/2) + 5n = 2 \cdot (2T(n/4) + 5 \cdot \frac{n}{2}) + 5n = 4T(n/4) + 2.5n \\ &= 4(2T(n/8) + 5 \cdot \frac{n}{4}) + 2.5n = 8T(n/8) + 3.5n = \dots \\ &= n \cdot 0 + \log n \cdot 5n = 5n \log n \\ &\Rightarrow T(n) \neq O(n) \end{aligned}$$

Theory recap

Maximum subarray sum

Given: An array of numbers $a_1, \dots, a_n \in \mathbb{Z}$ (in this order)

Wanted: Maximum subarray sum, where a subarray is a sequence of consecutive elements in the array (we cannot skip elements)

$$S_{i,j} = a_i + \dots + a_j, \quad i, j \in \{1 \dots n\}, i \leq j$$

($S = 0$ if all elements of the array < 0 , this is the empty subarray)

Algorithm 1 (naiv)

Idea: Calculate the sum of all possible subarrays and choose the biggest one of them

Algorithm 1 ($a_1, \dots, a_n$)

for $i = 1, \dots, n$ do

for $j = i, \dots, n$ do

$S_{ij} := 0$ #initialize the sum of subarray i, j to 0

for $k = i, \dots, j$ do

$S_{ij} += a_k$

choose max from all S

Runtime analysis:

1) To calculate the sum $S_{ij} = \sum_{k=i}^j a_k$, we need $j-i$ additions
 $\Rightarrow$ the number of additions to calculate S_{ij} does not depend on j and i , but on their difference. This means that the number of additions we need in order to calculate a subarray sum is independent of the starting and ending index of the subarray. It only depends on the length of the subarray.

2) Now we are going to calculate how many arrays of each length we have.

length 1: starting index can be $1, 2, \dots, n \Rightarrow n$ arrays

length 2: starting index can be $1, 2, \dots, n-1 \Rightarrow n-1$ arrays

$\vdots$

length i : starting index can be $1, 2, \dots, n-i+1 \Rightarrow n-i+1$ arrays

$\vdots$

length n : starting index can be only 1 $\Rightarrow 1$ array

$\Rightarrow$ Total number additions:

$$\sum_{i=1}^n \underbrace{(n-i+1)}_{\text{number of subarrays of length } i} \cdot \underbrace{(i-1)}_{\text{number of additions}} = n \cdot 0 + (n-1) \cdot 1 + (n-2) \cdot 2 + \dots + 1 \cdot (n-1) =$$

$$\begin{aligned}
&= 1 + 1 + \dots + 1 \\
&\quad + 2 + 2 + \dots + 2 \\
&\quad + 3 + 3 + \dots + 3 \\
&\quad + 4 + \dots + 4 \\
&\quad \vdots \\
&\quad \vdots \\
&\quad + (n-1)
\end{aligned}$$

$$\geq \frac{1}{2} (1^2 + 2^2 + \dots + (n-1)^2) = \frac{1}{6} (n-1) \left(n - \frac{1}{2}\right) n \geq \frac{n^3}{24} \Rightarrow A(n) \geq O(n^3)$$

$$\leq \frac{1}{2} (n^2 + (n-1)^2 + \dots + 1^2) = \frac{1}{6} n \left(n + \frac{1}{2}\right) (n+1) \leq n^3 \Rightarrow A(n) \leq O(n^3)$$

$\Rightarrow A(n) = \Theta(n^3)$, where $A(n)$ denotes the number of additions for an input of length (n)

Algorithm 2

We notice:

$$\begin{aligned}
S_{i,i} &= a_i \\
S_{i,i+1} &= S_{i,i} + a_{i+1} \\
S_{i,i+2} &= S_{i,i+1} + a_{i+2} \\
&\vdots
\end{aligned}$$

$$S_{i,n} = S_{i,n-1} + a_n$$

Instead of computing $S_{i,j}$ from scratch, we can use the fact that we have already computed $S_{i,j-1}$.

This means that we can compute the sums of all subarrays starting from i with $n-i$ additions

$$\Rightarrow \text{Total number of additions} = \sum_{i=1}^n (n-i) = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \Theta(n^2)$$

Algorithm 3 (recursive, divide and conquer)

Idea: divide in two halves, for each half find the maximum subarray sum. Be careful with the subarrays that get divided by the split



Case 1:

max Sum

Case 2:

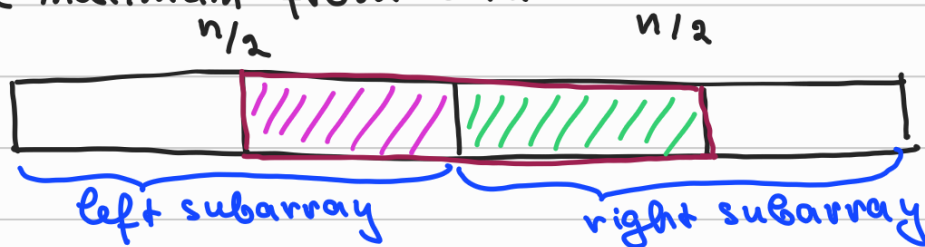
max Sum

Case 3:

max	Sum
-----	-----

In case 1 and 2, where the subarray with maximum sum is not split, we solve recursively for the left and right part and take the maximum from both.

Case 3:



subarray with maximum sum

// - this part of the subarray with maximum sum is the biggest suffix sum of the left subarray

// - this part of the subarray with maximum sum is the biggest prefix sum of the right subarray

Runtime analysis: Let $T(n)$ be the number of additions the recursion perform on an input of size n

$$\Rightarrow T(n) = \underbrace{2T(n/2)}_{\substack{\downarrow \\ \text{recursive} \\ \text{call for} \\ \text{left and} \\ \text{right subarrays}}} + \underbrace{O(n)}_{\substack{\text{linear time} \\ \text{to calculate} \\ \text{bigges suffix/prefix}}}$$

$$T(1) = C$$

$$\begin{aligned}\Rightarrow T(n) &= 2T\left(\frac{n}{2}\right) + O(n) = 2\left(2T\left(\frac{n}{4}\right) + O\left(\frac{n}{2}\right)\right) + O(n) \\ &= 4T\left(\frac{n}{2}\right) + O(n) + O(n) = 4T\left(\frac{n}{2}\right) + 2O(n) = \dots = \\ &= n \cdot T(1) + \log n \cdot O(n) \\ &= n \cdot C + \log n \cdot n \\ &= n \log n + C \cdot n\end{aligned}$$

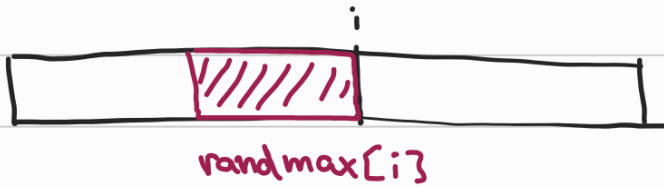
$$\Rightarrow \text{Total runtime } O(n \log n)$$

Algorithm 4:

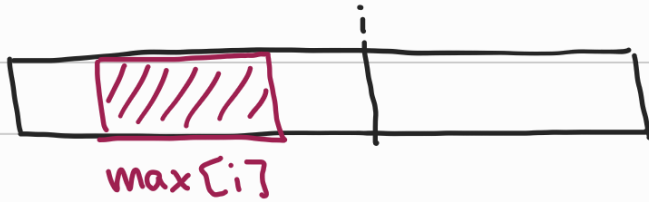
Idea: We define 2 sums:

randmax and max , where

$\text{randmax}[i]$ is the maximum subarray sum of the subarrays that end in index i



$\text{max}[i]$ is the maximum subarray sum out of all subarrays that end in index $\leq i$



! It is important to note that $\exists j \in [n]$, such that $\text{max}[j] = \text{rand}[j]$ unless all elements of the array are negative

$$\text{randmax}[i] = \max(a_i, \text{randmax}[i-1] + a_i), \text{randmax}[1] = a_1$$

$$\text{max}[i] = \max(\text{randmax}[i], \text{max}[i-1])$$

The correct answer is in $\text{max}[n]$

We calculate $\text{randmax}[i]$ and $\text{max}[i]$ $\forall i \in [n]$ with 1 traversal of the array

$\Rightarrow$ runtime $O(n)$